

ChessLab: Evaluating the Wisdom of Artificial Crowds in Chess

Author #11

Abstract

Wisdom of crowds suggests that aggregated decisions of diverse, independent agents often outperform individual experts. However, crowds of casual chess players still struggle against strong opponents. Recent projects show that the method by which a move is chosen by the crowd can significantly influence their performance. This work investigates whether artificial crowds can exhibit collective intelligence phenomena in chess. Key questions include: (1) Can voting ensembles of weak agents defeat stronger opponents? (2) How does diversity composition affect collective performance? (3) What voting mechanisms optimally aggregate engine decisions? We introduce ChessLab, an open-source framework that replaces human crowds with heterogeneous ensembles of deterministic and learning-based chess engines. The system enables reproducible experiments with detailed move-level evaluations and robust Elo estimation across diverse agent types on a unified scale. We validate this methodology through Elo coherence experiments, establishing MadChess as a practical reference standard due to its wide Elo range and reliable self-consistency. Preliminary experiments show that majority voting among diverse weak engines does outperform significantly the mean level of those engines. This paper presents the framework, experimental methodology, and a research roadmap bridging empirical chess results and broader theories of collective intelligence.

1 Introduction

Collective intelligence can in some circumstances exceed individual capability. One of the earliest demonstrations comes from Galton’s famous ox-weighting experiment, where the median guess of approximately 787 individuals in a county fair was more accurate than expert estimators and nearly matched the actual weight of 1,197 pounds [Galton, 1907]. This phenomenon has been tested in a variety of settings, such as prediction markets, where Atanasov et al. compared prediction markets with prediction polls in a large geopolitical forecasting tournament and found that carefully aggregated

team forecasts could outperform market prices [Atanasov et al., 2017]. Question-answering game shows also provide natural experiments.

In “Who Wants to Be a Millionaire”, the “Ask the Audience” lifeline yields correct answers around 87–92% of the time, whereas phone-a-friend experts succeed only about 55–65% of the time, across many national editions of the show [Surowiecki, 2004].

However, in competitive games like chess, crowds have historically struggled against strong opponents. The 1999 Kasparov vs. The World match saw more than 50,000 human players collectively challenge the World Chess Champion Garry Kasparov [Kasparov, 1999], reaching move 62 before conceding. More recently, in May 2025, over 143,000 players held World Champion Magnus Carlsen to a draw in a record-breaking game on Chess.com [Chess.com, 2025b], and in November 2025, over 200,000 players lost in a match against International Master Levy Rozman (Gotham-Chess) [Chess.com, 2025a]. However, these matches remain largely anecdotal: a single game provides minimal evidence for systematic collective intelligence, as outcome variance is high and learning effects are confounded.

A Fouloscopie project, led by Mehdi Moussaïd at the Max Planck Institute for Human Development, addressed this limitation by conducting one of the first rigorous, large-scale experiments on collective chess intelligence [Moussaïd and Garnier, 2022]. Approximately 25,000 participants were split into teams to play a total of 500 chess games against AI opponents. Using majority-vote aggregation of player moves, Moussaïd and colleagues demonstrated that crowds achieved a performance level similar to a master while being composed predominantly of beginners and casual players. Importantly, by controlling for poll visibility, the results showed a significant effect of social influence on the crowd’s level.

Yet even this landmark study leaves open critical questions:

- **Mechanism understanding:** What aspects of diversity (in skill, size, architecture) drive the ensemble advantage?
- **Optimization:** Can we identify decision-making processes that outperform simple majority aggregation?
- **Domain generality:** Does collective intelligence in chess rely on human-specific cognitive mechanisms, or do the principles apply to artificial agents as well?

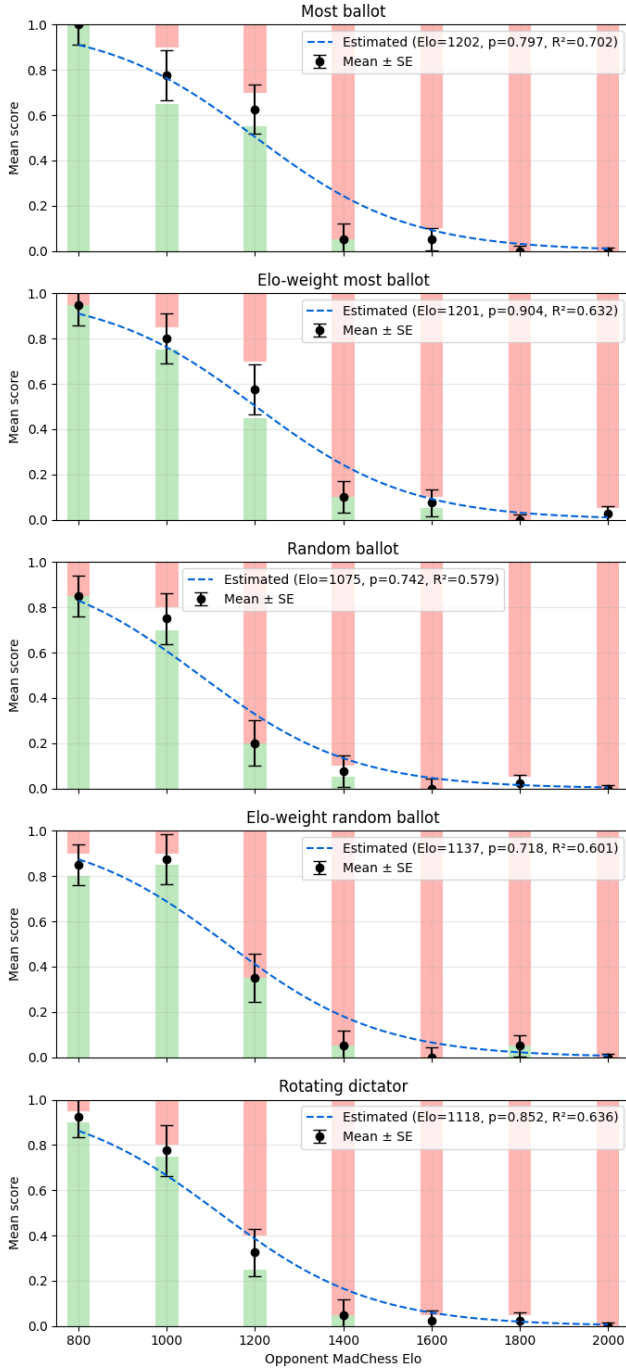


Figure 1: Performance of a MadChess crowd with variations on aggregation methods

We address these questions through **ChessLab**, an open-source framework for studying artificial collective intelligence in chess. By substituting human crowds with heterogeneous ensembles of chess engines, we gain precise control over experimental conditions, perfect reproducibility, and the ability to run experiments rapidly with modest computational cost.

The main contributions of this paper are:

- Framework and methodology:** A modular, extensible system for running ensemble chess experiments at scale, with persistent storage of games, moves, evaluations, and statistical analyses.
- Theoretical grounding:** Clear hypotheses derived from Scott Page’s Diversity Prediction Theorem and classical wisdom-of-crowds theory, adapted to the chess domain.
- Comprehensive research roadmap:** A structured plan for experiments addressing collective intelligence across different engine types, ensemble compositions, and voting strategies.

This paper focuses on framework description, experimental design, and hypothesis articulation. Results from additional experiments will be presented in subsequent publications as they are completed.

2 Related Work

2.1 Collective Intelligence and Wisdom of Crowds

The wisdom of crowds phenomenon emerged from early empirical observations: Galton’s ox-weighing experiment demonstrated that collective judgments (median of 787 guesses) were more accurate than nearly all individual estimates [Galton, 1907]. Surowiecki systematized this observation, arguing that under certain conditions (diversity of opinion, independence of members, decentralization, and effective aggregation) groups reliably outperform experts [Surowiecki, 2004]. Formally, Scott Page’s *Diversity Prediction Theorem* provides mathematical grounding:

$$\text{Collective Error} = \text{Individual Error} - \text{Diversity} \quad (1)$$

where collective error is the mean squared error of the group prediction, Individual Error is the average of individual squared errors, and Diversity captures how different agents’ predictions are [Page, 2007]. This result implies that a team of ordinary individuals can outperform a specialist, provided the group is large enough and exhibits sufficient diversity in cognitive approach.

However, diversity alone is insufficient. Studies show that social influence, information cascades, and herding can destroy the diversity necessary for wisdom of crowds [Surowiecki, 2004; Almaatouq *et al.*, 2020]. Almaatouq *et al.* demonstrated that agents strategically choosing whom to learn from (adaptive networks) better maintain collective intelligence than static networks [Almaatouq *et al.*, 2020]. Koriati and colleagues showed that *confidence* acts as a metacognitive signal: when individuals can estimate how well-calibrated their judgment is, confidence-weighted aggregation outperforms simple majority voting [Koriati, 2011], except when facing common misconceptions that result in misplaced confidence. Prelec extended this with the “surprisingly popular” voting method, which selects answers that are more popular than subjects predicted, revealing hidden consensus even when the majority is wrong [Prelec *et al.*, 2017].

In summary, collective intelligence requires: (1) a sufficient number of participants, (2) diversity in decision-making, (3) independence of judgments, (4) appropriate ag-

gregation mechanisms, and possibly (5) calibration of confidence when weighting is available.

2.2 Collective Intelligence in Chess

Chess provides an ideal laboratory for studying collective decision-making in complex, sequential tasks. Unlike static judgment tasks (trivia answering, estimation), chess involves:

- **High-dimensional state space:** Approximately 10^{47} legal positions, making expert heuristics difficult and providing an extensive test dataset.
- **Strategic thinking:** Each player must evaluate not just the next move but plan for responses and counter-responses. This task allows deliberation to express emerging strategies.
- **Objective evaluation:** The outcome of a game is unambiguous, and thanks to modern engines, each move can be evaluated with precise numerical scoring. The performance level of a crowd can thus be estimated with high precision.
- **Cultural significance:** Chess remains a synecdoche for intelligence. Both humans and algorithms are expected to demonstrate competence at this task, providing abundant documentation and research literacy.

To evaluate the level of a chess player the community mainly relies on a number called the Elo rating. Given two players with Elo ratings Elo_A and Elo_B , the expected score (probability of A winning) is:

$$E_A = \frac{1}{1 + 10^{(Elo_B - Elo_A)/400}} \quad (2)$$

This is the standard Elo formula, calibrated for chess [Elo, 1978]. While this metric captures relative strength between two players, its absolute interpretation depends heavily on methodology and context. Community conventions suggest that players understanding the basic rules rate between 600 and 900 Elo, casual players typically orbit around 1300 Elo, and from 2000 Elo onward, players enter the master and grandmaster territory [Elo, 1978]. At his peak in May 2014, Magnus Carlsen achieved a classical rating of 2882, the highest in history, and he also achieved 2909 in the newly introduced Freestyle Chess (Chess960) rating system [FIDE, 2025].

2.3 Empirical Studies of Crowd Chess

The Kasparov vs. The World match (1999) is the earliest documented crowdsourced chess challenge. Over four months, approximately 50,000 amateur and intermediate players voted on moves against the then-World Champion. Despite reaching an advanced position, the crowd eventually conceded after 62 moves when facing maximum resistance from Kasparov [Kasparov, 1999]. While historically significant, this single-game format provides weak evidence: a single outcome cannot be reliably attributed to collective intelligence versus chance.

The Fouloscopie experiment rectified this limitation through large-scale repetition [Moussaïd and Garnier, 2022]. Moussaïd and colleagues organized approximately 25,000

human participants to play a total of 500 games against chess engine opponents. Key design features included:

- **Diverse player population:** Participants self-reported their Elo rating; the population was approximately normally distributed with a mean around 1165 Elo.
- **Majority-vote aggregation:** Players had several hours to evaluate positions and vote on the next move; the most-voted move was played automatically.
- **Opponent variety:** Chess engine opponents (Maia models) ranged from weak (1100 Elo) to strong (1900 Elo), allowing measurement of performance across the strength spectrum.
- **Social influence manipulation:** Comparing voting-only versus poll-visibility conditions revealed how information visibility affects collective decision quality.

Results showed that crowds achieved a win rate exceeding 60% across all tested opponents. With poll visibility, win rates improved to approximately 77%, while visibility-hidden conditions showed approximately 64% win rates. Importantly, the crowd played measurably better than the average individual member, confirming classic wisdom-of-crowds predictions.

2.4 Chess Engines

Modern chess engines span multiple paradigms, each with distinct decision-making processes. We focus on three particularly relevant approaches.

Stockfish

Stockfish, as of 2025, is among the world’s strongest chess engines. It is an open-source tree-search algorithm combining classical techniques (alpha-beta pruning, endgame tables) with small neural networks for position evaluation [Stockfish Development Team, 2024]. Its strength is calibrated via Elo ratings: users can set `UCI_Elo` to values in an approximate range from 1320 to 3190. The engine deliberately weakens its play through stochastic evaluation perturbation (seeded random noise), ensuring exploration of different move trees at reduced strength levels.

Maia: Human-Mimetic Neural Networks

In contrast to optimal play, Maia learns to *predict human moves* from millions of Lichess games [McIlroy-Young *et al.*, 2020]. Rather than computing optimal moves, Maia generates a probability distribution over legal moves, parameterized by player skill level. Maia-2, introduced as a follow-up architecture, incorporates a skill-aware attention mechanism that dynamically integrates player skill level with board position encoding [Tang *et al.*, 2024]. This mechanism achieves approximately 50–52% top-1 move accuracy across human players rated 1100 to 1900 Elo. In practical play, Maia replicates human decision-making nearly indistinguishably, making it an excellent tool for evaluating crowd performance against human-like opponents.

Mad Chess

MadChess is a classical alpha-beta search engine similar to Stockfish. Written in C#, its latest release (v3.3) reaches a

playing strength above 2800 Elo in its strongest configuration and can easily perform over 2400 even cap at a fraction of a second per move [Madsen,]. What makes MadChess particularly attractive for experiments requiring a wide and realistically calibrated Elo spectrum, is its strength reduction algorithm. With parameters explicitly tuned for “realistic” weakened play, it mimic player with precise level going as low as 600 Elo. We leverage theses capabilities in our ensemble experiments (Section 4.1).

Large Language Models

Recent work demonstrates that Large Language Models (LLMs) trained on chess game transcriptions can develop emergent internal representations of board states [Karvonen, 2024]. These representations can be altered, thereby changing the move predicted by the LLM. This represents a fundamentally distinct paradigm: LLMs make decisions based on statistical pattern recognition from training data rather than explicit search or neural evaluation. Their move quality is unstable and depends on many factors (model architecture, PGN-based prompting, board representation, move history), which are often difficult to replicate or interpret.

Prelec’s “single-question crowd wisdom” result [Prelec *et al.*, 2017] and related work suggest that aggregating judgments with access to meta-predictions or rationales can be especially powerful, which motivates using LLMs as both players and commentators in chess ensembles.

But recent work such as LLM CHESS introduces a dynamic benchmark that evaluates reasoning and instruction-following in LLMs via full chess games against configurable opponents, reporting that even state-of-the-art reasoning models often struggle to reliably win or complete games. [Kolasani *et al.*, 2025] Carlini showed that `gpt-3.5-turbo-instruct` can achieve a playing strength of roughly 1788 Elo under specific conditions [Carlini, 2023]. More recent transformer-based models have achieved grandmaster-level performance when trained directly on game transcriptions [Ruoss *et al.*, 2024].

Although modern chat-oriented LLMs appear to lag behind specialized models, Dynomight [Dynomight, 2024] demonstrated that prompting strategies (such as requiring the model to regenerate the move history before outputting the next move) can dramatically improve their move quality. These results suggest that apparent “unusual” chess behavior in some LLMs may stem less from intrinsic chess competence and more from specific training setups and prompt configurations. For ensemble experiments, LLMs offer substantial cognitive diversity and potential for open deliberation, but also present engineering challenges that must be carefully controlled (API management, prompt optimization, legal move filtering).

2.5 Aggregating Engines

Carvalho *et al.* [Carvalho *et al.*, 2017] studied majority voting among homogeneous groups of checkers engines. They found that group performance improves roughly logarithmically with group size, while outcome variance decreases. In chess, Spoerer *et al.* [Spoerer *et al.*, 1999] investigated three-member simple majority voting among engines of varying strength. They analyzed how different configurations

(e.g., equal-strength engines versus mixtures of stronger and weaker ones) affect ensemble performance and the variance of game outcomes. Their results indicate that even very small committees can benefit from majority aggregation, but that gains depend sensitively on the diversity and relative strength of the members, as well as on how ties and disagreements are resolved. According to this study, to observe a stronger wisdom-of-crowds effect, one should focus on low-Elo engines with small strength differences, which seems to contradict the diversity assumption.

Another team have explicitly optimized engines for *skill-compatibility* in collaborative chess settings, where near-optimal engines are adapted to cooperate effectively with much weaker partners rather than simply maximizing standalone performance [Hamade *et al.*, 2024].

3 Methodology and Framework

ChessLab is a modular, open-source framework for large-scale chess engine evaluation and crowd experimentation. In this section, we detail some of its inner workings and base modules.

3.1 UCI Engines

Chess engines communicate with an interface named the Universal Chess Interface (UCI), an open text-based protocol that standardizes how engines and user interfaces exchange commands and options, and is now supported by most modern chess programs [Meyer-Kahlen and Huber, 2000].

We leverage this protocol to integrate any UCI-compliant engine, such as Stockfish. For engines or models that do not expose UCI (for example, Maia), we provide a custom wrapper interface. In addition, we implement several synthetic agents internally (e.g., random baselines and LLM-based agents). By exposing all engines through a common abstraction, we can treat them uniformly and combine them into a “crowd” engine governed by a configurable aggregation rule.

3.2 Aggregator Rules

We define a family of aggregation rules that map a set of engine proposals into a single move choice. Examples include:

- **Majority:** Each engine proposes a single move; the move with the highest vote count is played, with ties broken randomly.
- **Minority:** The least popular move among the proposed ones is selected, intended as an extreme contrarian baseline.
- **Randomized:** One of the proposed moves is sampled at random, with or without weighting by engine Elo.
- **Top-Elo dictator:** The move proposed by the highest-rated engine in the ensemble is always selected.
- **Bottom-Elo dictator:** Symmetric to the top-Elo dictator, using the lowest-rated engine; serves as a control for “anti-expert” aggregation.
- **Median-Elo dictator:** The move from the median-rated engine is chosen, approximating a representative member of the group.

- **Rotating dictator:** Engines take turns acting as dictator across moves or games, ensuring equal participation over time.

- **Elo-weighted:** Each engine casts a vote weighted by a function of its Elo (e.g., proportional to Elo, or logistic in Elo); the move with the highest total weight is selected.

These rules are not intended to lead to optimal results, but rather to be compared against each other and to test their effectiveness in artificial crowds relative to human crowds. Their formalism come from the voting and preference aggregation studies of computational social choice sciences [Brandt *et al.*, 2016].

3.3 Game Execution Engine

The core of ChessLab is a Python-based game runner built on top of the `python-chess` library, which manages internal move generation and game-state transitions. Games are executed asynchronously using `asyncio`, allowing highly parallelized evaluation across many engine-versus-engine matchups. The PostgreSQL schema is organized around the following core entities:

- **Players:** Engine type (e.g., Stockfish, Maia, LLM, random), name, nominal Elo rating, creation timestamp, and configuration parameters such as depth limits or temperature settings.
- **Games:** References to white and black player IDs, the numeric result (1–0, 0–1, 1/2–1/2), full PGN, and meta-data about the game.
- **Moves:** For each game, a sequence of moves with SAN and UCI representations and the FEN preceding each move.
- **Evaluations:** For a given move, the required engines (player engine or multiple engines in case of a crowd) produce an evaluation (in centipawns or similar score).

3.4 Statistical Analysis Module

ChessLab evaluates engine and ensemble performance through statistical analysis at multiple levels of granularity: individual games, pairwise matches, and multi-opponent campaigns. The framework implements hypothesis testing procedures to distinguish genuine performance differences from random variation, and it provides Elo rating estimation with uncertainty quantification.

Per-Game Metrics

For each completed game, ChessLab computes both outcome-based and quality-based metrics. The outcome is encoded as a score $s \in \{1, 0.5, 0\}$ representing win, draw, or loss respectively. Beyond win/loss statistics, we evaluate move quality through centipawn loss analysis, where each move’s quality is assessed by comparing the position evaluation before and after the move using Stockfish at depth 10. For a player who made N moves in a game, the average centipawn loss is:

$$\text{ACPL} = \frac{1}{N} \sum_{i=1}^N |\text{eval}_{\text{before}_i} - \text{eval}_{\text{after}_i}| \quad (3)$$

where evaluations are reported from the moving player’s perspective. This metric provides an independent assessment of play quality that complements outcome-based statistics, particularly useful for fast evaluation on specific positions.

Elo Rating Estimation

ChessLab estimates Elo ratings using the standard logistic model.

Single-Opponent Estimation. When a player achieves mean score $\bar{s}$ over n games against a single opponent with known rating R_{opp} , we invert equation (2) to obtain:

$$\hat{R} = R_{\text{opp}} + 400 \log_{10} \left(\frac{\bar{s}}{1 - \bar{s}} \right) \quad (4)$$

To avoid numerical issues at the boundaries, we clip observed scores to the range $[\epsilon, 1 - \epsilon]$ where $\epsilon = 10^{-9}$, and constrain the final estimate to $[250, 3000]$ Elo.

Multiple-Opponent Estimation. When evaluating against m opponents with known ratings $R_1, \dots, R_m$ and corresponding observed scores $\bar{s}_1, \dots, \bar{s}_m$ from $n_1, \dots, n_m$ games, we estimate the player’s Elo by minimizing the weighted squared error between observed and expected scores:

$$\hat{R} = \arg \min_{R \in [250, 3000]} \sum_{i=1}^m w_i (\bar{s}_i - E(R, R_i))^2 \quad (5)$$

where the weights are normalized as $w_i = n_i / \sum_{j=1}^m n_j$. This objective function is optimized using bounded scalar minimization with a tolerance of 0.1 Elo points. The resulting estimate $\hat{R}$ represents the single rating that best explains the ensemble of results across all opponents, accounting for both the strength of each opponent and the number of games played.

Statistical Testing

ChessLab performs hypothesis testing at two levels: single matches against one opponent and ensemble evaluation across multiple opponents. Both use the same underlying statistical framework but differ in how mean scores and standard errors are computed.

Single-Match Analysis. A match consists of n games between two fixed players. Given individual game scores $s_1, s_2, \dots, s_n$, the mean score is:

$$\bar{s} = \frac{1}{n} \sum_{i=1}^n s_i \quad (6)$$

For an expected score p (derived from Elo ratings via equation (2)), the standard error is:

$$\text{SE}(p) = \sqrt{\frac{p(1-p)}{n}} \quad (7)$$

Multi-Opponent Ensemble Analysis. When evaluating against m opponents with Elo ratings $R_1, R_2, \dots, R_m$, where match i consists of n_i games with observed mean score $\bar{s}_i$ and

expected score p_i , we compute weighted ensemble statistics. The ensemble observed mean score is:

$$\bar{s}_{\text{ensemble}} = \frac{\sum_{i=1}^m n_i \bar{s}_i}{N}, \quad \text{where } N = \sum_{i=1}^m n_i \quad (8)$$

The ensemble expected mean is computed analogously:

$$\bar{p}_{\text{ensemble}} = \frac{\sum_{i=1}^m n_i p_i}{N} \quad (9)$$

To account for heterogeneous expected scores across matches, the standard error of the pooled proportion is:

$$\text{SE}_{\text{ensemble}} = \frac{\sqrt{\sum_{i=1}^m n_i p_i (1 - p_i)}}{N} \quad (10)$$

This formulation correctly handles cases where different matches have different expected outcomes, which is essential when testing an engine against a ladder of opponents at varying strength levels.

Hypothesis Testing. For both single-match and ensemble analyses, the test statistic is:

$$z = \frac{|\bar{s} - p|}{\text{SE}(p)}, \quad (11)$$

where $\bar{s}$ and p represent either the single-match or ensemble values as appropriate. We automatically select between z-test and t-test based on sample size and expected frequency. For single matches, when $np \geq 5$ and $n(1 - p) \geq 5$, we use the normal approximation (z-test); otherwise we use the t-distribution with $n - 1$ degrees of freedom. For ensemble analysis, we apply a conservative check: if $\min_i(n_i p_i) \geq 5$ and $\min_i(n_i(1 - p_i)) \geq 5$, we use the z-test; otherwise we use a t-distribution with $N - m$ degrees of freedom. The two-tailed p-value quantifies the probability of observing results as extreme as $\bar{s}$ under the null hypothesis that the player’s true strength matches expectation. We use this testing framework throughout ChessLab to assign estimated Elo ratings across heterogeneous agent types on a unified scale.

4 Proof-of-Concept Experiments

In this section we present preliminary experiments designed to validate ChessLab’s Elo calibration procedures and demonstrate basic ensemble behaviors. Since there is no absolute standard for engine Elo ratings, we first test whether engines are internally coherent with respect to their own Elo settings. We then establish a practical reference standard for cross-engine comparisons and demonstrate collective intelligence phenomena. These results serve as methodological validation and motivate the more extensive roadmap outlined in Section 5.

The figures are auto-generated by the statistical analysis module. Columns represent the measured scores obtained in practice across different strength levels. Green represents the victory ratio, red the loss ratio, and white the draw ratio. Curves, with a specified Elo, represent the expected mean score (abscissa) for a player of that Elo against an opponent of a given Elo (ordinate).

4.1 Elo Coherence Experiments

To establish a reliable foundation for comparing engines across different architectures, we evaluate the internal Elo coherence of three diverse engines:

Stockfish. Stockfish demonstrates strong internal coherence when configured at depth 10. Across the tested strength levels, the estimated Elo values align closely with the nominal calibration, as shown in Figure 3 in Appendix ??.

MadChess. Similarly in Figure 2, MadChess exhibits robust Elo coherence across its exceptionally wide rating range (600–2600 Elo) when configured at 0.3 seconds per move. This broad coverage makes MadChess particularly valuable for testing weak agents and diverse ensembles.

Maia2. In contrast, Maia2 displays less consistent Elo coherence across its supported range (1100–1900 Elo) when run without explicit time constraints, as shown in Figure 4 in Appendix ??. This finding may tentatively indicate a potential issue with the implementation or calibration suggested by the original authors, though we note that the Maia team is actively developing Maia3 with wider Elo coverage. Further investigation is needed to determine whether these deviations stem from our experimental setup or the engine’s intrinsic characteristics.

Cross-Engine Validation. To verify that different engines’ Elo scales are mutually consistent, we conducted head-to-head matches between Stockfish and MadChess. The results, provided in Figure 5 in Appendix ??, show some agreement between their rating systems. Stockfish seems to be underperforming by up to 200 Elo in lower lever but getting closer by to the expectation in higher level. Given this alignment, combined with MadChess’s speed, superior Elo range and demonstrated coherence, we adopt MadChess as our practical reference standard for subsequent experiments. This choice enables testing of very weak agents while maintaining reliable strength calibration across the full rating spectrum.

4.2 Crowd Ensembles

Having established MadChess as our reference standard, we construct an ensemble of 10 MadChess instances with Gaussian-distributed strength (mean 1100 Elo, standard deviation 200 Elo) and evaluate various aggregation rules against a ladder of seven MadChess opponents spanning 800–2000 Elo. To permit rapid iteration, we limit this proof-of-concept to 20 games per opponent.

Results are illustrated in Figure 1. Preliminary observations suggest that majority voting with its estimated 1202 Elo can significantly outperform the average individual member (1048 Elo for this experiment). Other aggregation methods (randomized ballot, Elo-weighted variants, rotating dictator) yield similar though less performant profiles.

These results suggest that even simple majority voting among a small but diverse group of weaker engines can achieve collective intelligence gains in this domain. The question is now how much gain can be leverage.

4.3 Local LLM

Finally, we test three local LLMs with naive prompting:

552	• Meta Llama-3.2-1B-Instruct	604
553	• Qwen Qwen2.5-1.5B-Instruct	605
554	• Google Gemma-3-1B-IT	606
555	Board states were encoded in plain algebraic notation (PAN)	
556	with the prompt “Given the following chess position, what is	
557	the best legal move? ”. In this setting, the LLM agents failed	
558	to win a single game across all opponents, and average loss	
559	per move exceeded 1800 centipawns. These results highlight	
560	both the potential and the challenges of incorporating LLM-	
561	based agents into ChessLab: substantial prompt engineering	
562	(e.g., move history reconstruction [Dynomight, 2024]), legal-	
563	ity filtering, and domain-specific fine-tuning are required be-	
564	fore LLMs can serve as a measurable and reliable member of	
565	the crowd.	
566	5 Roadmap	
567	The current work establishes ChessLab’s core infrastructure,	
568	validates its Elo calibration methodology through coherence	
569	experiments, and demonstrates basic ensemble feasibility.	
570	However, many key questions about artificial collective in-	
571	telligence in chess remain open. We outline here a roadmap	
572	of planned experiments and framework extensions.	
573	Planned experiments include:	
574	• Exploring the variation on engines, size and sampling	
575	distribution on the crowd ensembles experiments.	
576	• Calibrating distilled Stockfish and Maia variants to ob-	
577	tain lower-Elo opponents. Distilled engines use stochas-	
578	tic weakening mechanisms (e.g., random-move proba-	
579	bilities) to benchmark very weak chess engines [Mur-	
580	phy VII, 2019]. We also hope to propose a model of	
581	byzantine fault tolerance in chess playing crowd.	
582	• Reproducing the Moussaïd experiment as closely as pos-	
583	sible, using estimates of participant strength, and com-	
584	paring human and artificial crowds under analogous con-	
585	ditions [Moussaïd and Garnier, 2022].	
586	• Systematically exploring LLM prompting strategies	
587	(e.g., reconstructing move history, chain-of-thought rea-	
588	soning) to improve LLM move quality and stabil-	
589	ity [Dynomight, 2024].	
590	• For sufficiently capable LLMs, attempting to assign ef-	
591	fective Elo ratings and integrating them into mixed-	
592	architecture ensembles.	
593	• Having LLMs output their top- <i>k</i> candidate moves	
594	with rationales, enabling more sophisticated aggregation	
595	rules (e.g., Condorcet, Borda, Coombs, approval vot-	
596	ing).	
597	• Studying information-sharing regimes, such as letting	
598	LLMs observe current poll results before voting, to sim-	
599	ulate social influence dynamics.	
600	• Investigating argumentation pipelines in which multiple	
601	LLMs propose moves and arguments, and a separate “ar-	
602	bitrator” LLM selects the final move.	
603	Planned framework extensions include:	
	• Adding tactical puzzle modes to estimate Elo via cen-	604
	tipawn loss on curated positions, accelerating prelimi-	605
	nary testing on LLM-based engines.	606
	• Integrating external API-based LLMs, with rate-limit-	607
	aware scheduling in the request queue.	608
	• Providing containerized deployments (e.g., Docker im-	609
	ages) for reproducible local and cloud experiments.	610
	• Developing a web interface to share experiments with	611
	a broader audience and facilitate crowdsourced research	612
	contributions.	613
	6 Conclusion	614
	This paper introduces ChessLab, a framework for studying	
	collective intelligence in chess by replacing human crowds	
	with ensembles of artificial agents. The system enables hun-	
	dreds of reproducible, low-cost experiments that test hypothe-	
	ses from wisdom-of-crowds theory in a complex, sequen-	
	tial decision-making domain. By instrumenting games with	
	detailed move-level evaluations and robust Elo estimation,	
	ChessLab makes it possible to disentangle ensemble benefits	
	arising from diversity, bias correction, and variance reduc-	
	tion, and to compare artificial and human crowds on a com-	
	mon quantitative scale. The roadmap from baseline replica-	
	tion of human experiments to optimization of voting mecha-	
	nisms and information structures is designed to progressively	
	bridge empirical chess results and broader theories of collec-	
	tive intelligence.	
	We release ChessLab as open-source software and invite	
	the community to extend, replicate, and build upon this	
	work: https://github.com/Uspectacle/chesslab Future publi-	
	cations will report on the larger set of experiments outlined	
	here and on the resulting refinements to our understanding of	
	artificial and human crowd wisdom in strategic games.	
	Disclaimer	636
	The assistance of Large Language Models (LLMs) was used	
	in a limited and controlled manner during the writing process	
	of this paper. All scientific claims, experimental designs, the-	
	oretical contributions, and the final version of this paper re-	
	main the responsibility of the authors.	
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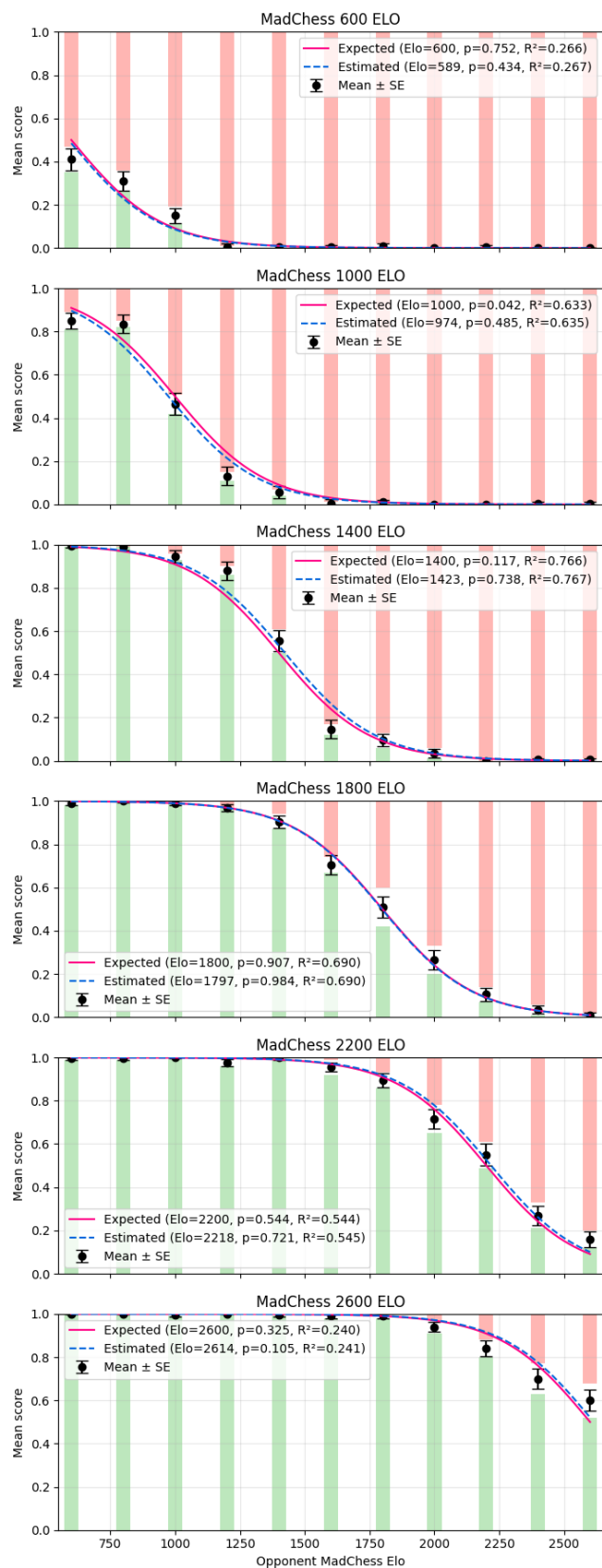


Figure 2: Elo estimation for a range of MadChess engines (0.3 sec-per move)

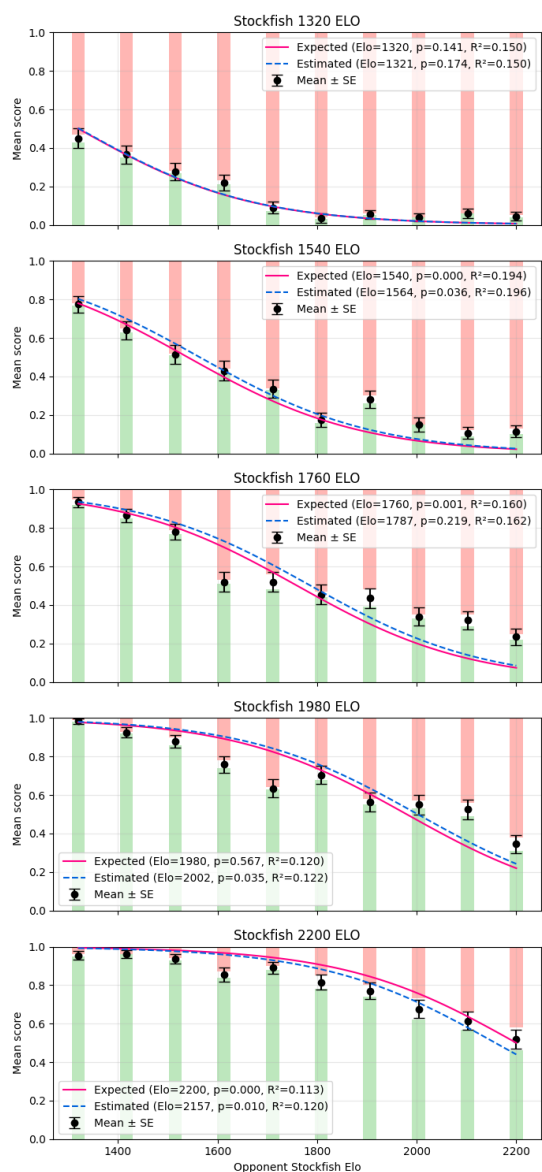


Figure 3: Elo estimation for a range of Stockfish engines (depth 10)

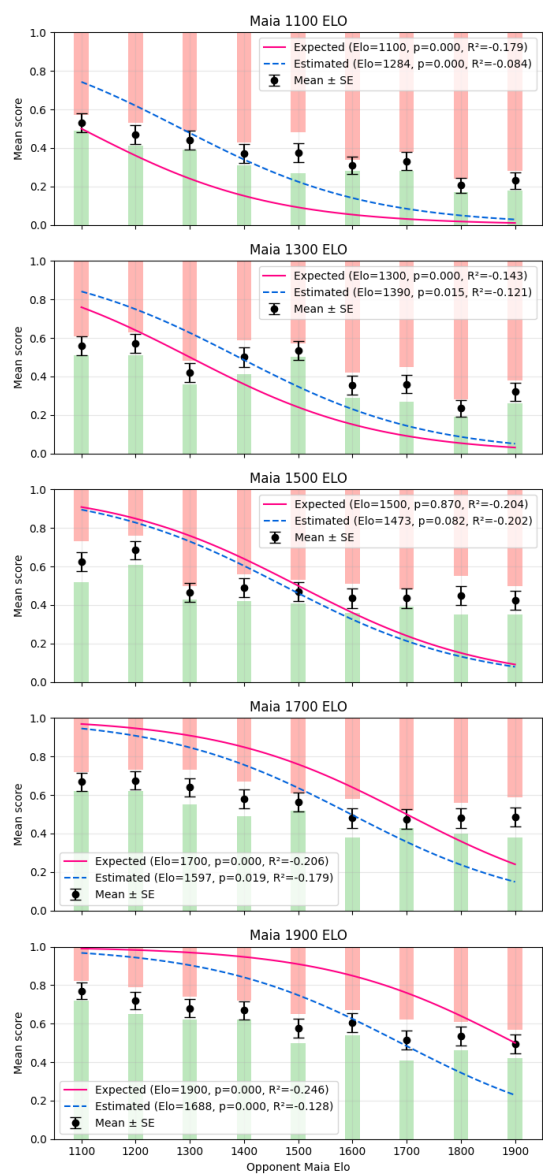


Figure 4: Elo estimation for a range of Maia engines

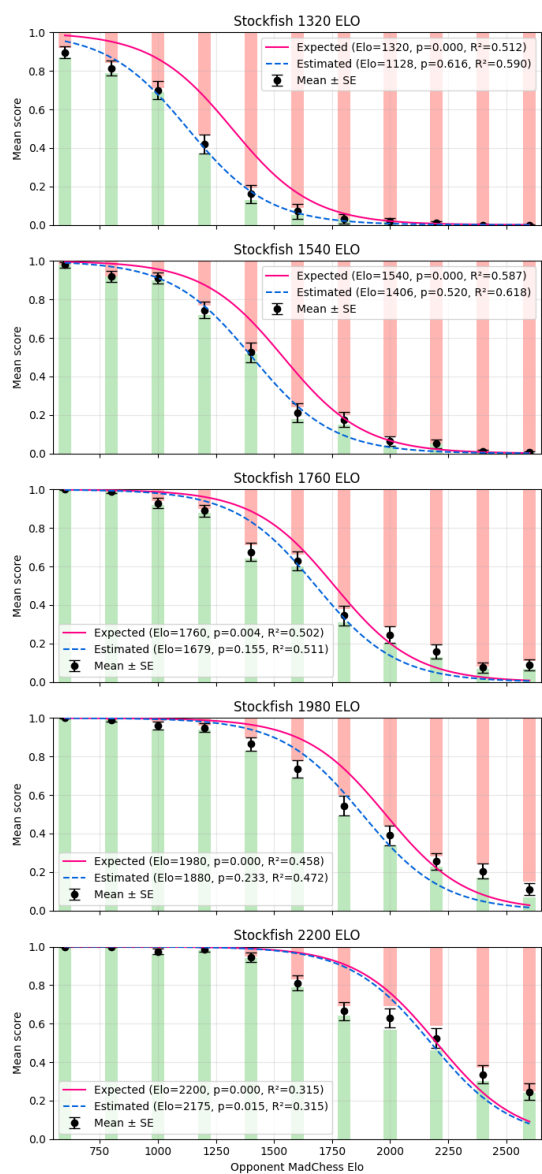


Figure 5: Elo estimation for cross-validation between Stockfish and MadChess